Forecasting small-business AI use six weeks ahead

ISYE 4031 Milestone 2 • Group 1 • October 2026

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1. Question and response to feedback

Question. Can a regression forecast an industry-by-size group's AI-use rate three Census releases (about six weeks) ahead more accurately than simple historical forecasts, and is it reliable enough to guide outreach planning? **Target:** the published percentage of employer businesses in a group that used AI in the last two weeks (Census BTOS). Small business means 1–99 employees. **Use:** deciding which industries and firm sizes to prioritize for AI outreach. The forecast describes groups, not firms, and says nothing about the effect of outreach.

Feedback and response.

- *TA, Milestone 1:* define the target and model, use several predictors, fit a baseline, and respect the short history. We state the model (Section 4), use current rate, industry, size, and trend, compare two naive forecasts, and evaluate only in time order.
- *October 20 meeting (relayed by Chris):* compare industries and show uncertainty. We added industry growth estimates with intervals (Figure 1) and separate regressions per industry.
- *October 27–28 (professor, relayed by Chris):* consider AI model-release timing, benchmark scores, and a nonstationary Poisson process. We modeled release arrivals and tested release features in the adoption forecasts (Section 5). The exact scope of this request is not yet confirmed.

2. Data and preparation

Source. Census Business Trends and Outlook Survey, sector-by-employment-size workbook, downloaded October 20, 2026 and hashed. One observation is a NAICS sector × size band in one biweekly release. Variables: AI-use rate and expected-use rate (%), the rate's standard error (SE, percentage points), sector, size band, release, and publication date. We use the 23 releases from the November 2025 question revision (202524) through release 202620, published October 8, 2026; earlier releases asked a narrower question.

Pipeline step	Count
Potential cells: 20 sectors × 5 size bands × 23 releases	2,300
Suppressed by Census, kept as missing (not imputed)	519 (22.6%)
Groups kept: rate and expected use in ≥7 of the first 8 releases	71 groups, 15 industries
Issue–target pairs ($t \to t{+}3$), complete cases	1,406
Scored forecasts: 10 issue dates, April 23 – August 27, 2026	706

Quality. No duplicate group-release keys; all rates lie in 0–100. Suppression is 20–25% in every size band. Survey noise rises with firm size: the median SE of a forecast target is 1.7 pp for 1–4 employees and 5.2 pp for 50–99. The 50–99 band keeps only 11 of 15 industries. An earlier third-party extract of the same data matches this workbook exactly on all 1,781 rates, 1,825 expected-use rates, 1,781 SEs, and missingness; one publication date differs by a day (release 202526) and does not affect any forecast.

Dependence and leakage. The same 71 groups recur every release, and a new BTOS sample year began with release 202616 (collected from July 27, 2026), so the 706 forecasts share 10 dates and common shocks; they are not independent. A forecast is issued on release t's publication date. Training uses only pairs whose target was already published; screening uses only the first 8 releases; preprocessing is fit on training rows. The code asserts these rules, and we confirmed them with an independent refit.

3. Exploratory findings

Estimated adoption rose in every industry, but at different speeds. Finance, Information, and Professional services grew about 1.1–1.3 pp per four weeks; Accommodation and food grew 0.16. Each of the first three appears in the bootstrap top three in ≥97% of draws, conditional on the model; their order is uncertain. This motivated separate industry models.

The second pattern is measurement uncertainty. Two SEs for a typical 50–99 estimate are about 10 pp, illustrating how survey noise can obscure changes. We compare forecast errors with the target's SE as context, not as a hard error floor (Figure 2).

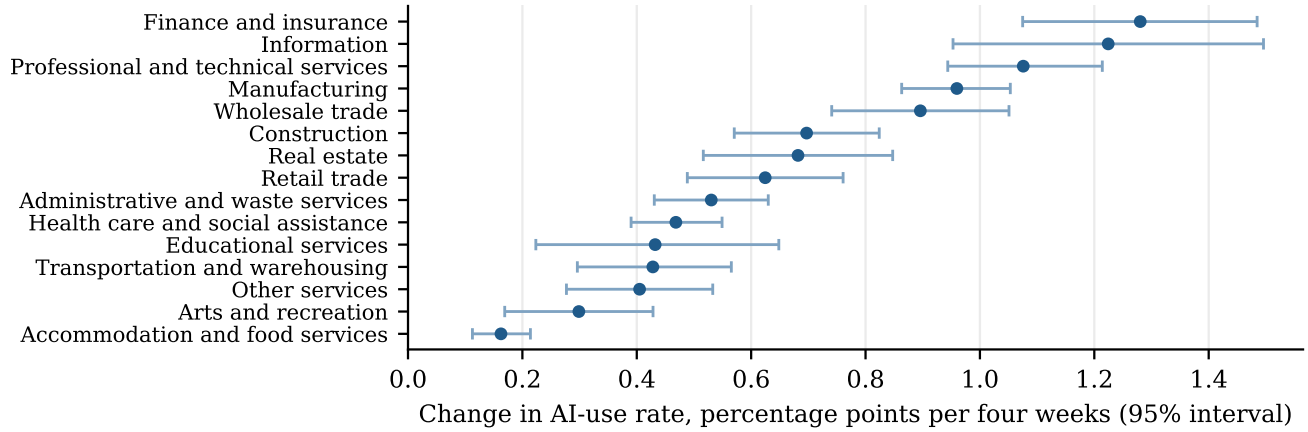


Figure 1. Estimated growth in AI use by industry, 71 groups, releases 202524–202620. Intervals: 95% moving-block bootstrap (2,000 draws). They describe this period and are conditional on the model; they are not forecasts.

4. Baseline and results

Baseline. Pooled ordinary least squares for group g at issue release t:

$$y_{g,t+3} = \beta_0 + \beta_1 y_{g,t} + \alpha_{\text{industry}} + \gamma_{\text{size}} + \beta_2 t + \varepsilon_{g,t+3}.$$

Every term is known at issue. **References:** carry the latest rate forward; mean of the group's latest six releases. **Extension:** the same equation fit separately within each industry. **Validation:** expanding window, refit at each of 10 issue dates, horizon three releases. **Metric:** mean absolute error (MAE) in pp; RMSE penalizes large misses. Differences are averaged over dates, with 95% moving-block bootstrap intervals over dates.

Forecast	MAE	RMSE	Bias	Change vs. six-release mean [95% CI]	Better at
Carry latest rate forward	4.15	5.75	−1.05	—	—
Six-release mean	3.39	4.64	−2.07	reference	—
Pooled regression (baseline)	**3.30**	**4.55**	+0.07	−0.09 [−0.22, +0.04]	7 of 10 dates
Separate industry regressions†	3.04	4.29	−0.21	−0.35 [−0.46, −0.24]	9 of 10
Gradient boosting†	3.08	4.33	−0.45	−0.31 [−0.42, −0.22]	10 of 10
Unregularized regression, all predictors†	6.01	8.15	+3.44	+2.61 [+0.30, +5.86]	1 of 10

Table 1. Out-of-sample error (pp), 706 forecasts at 10 issue dates. †Designed after these outcomes were published; a retrospective result, not a test.

Results. The baseline beats carrying the rate forward by 0.85 pp but beats the six-release mean by only 0.09 pp, and the interval includes zero. Its main gain is removing bias: the six-release mean lags a rising trend by 2.1 pp. In the original 9-date evaluation, the last fit's coefficient on the current rate is 0.21, so the model behaves like a smoothed group average plus a trend. Expected use added nothing (MAE 3.33 vs. 3.33 in the 9-date original evaluation). Separate industry models cut MAE by 0.35 pp (10%) against the six-release mean and 0.26 pp against the baseline, and win at 9 of 10 dates. Both their interval and boosting's exclude zero, but these exploratory comparisons follow model selection on observed outcomes and may overstate the gains.

Diagnostics (original 9-date evaluation). Residuals fan out as fitted rates rise. The four largest Cook's distances all belong to Information firms with 50–99 employees (residuals up to 27 pp), but none is highly influential (maximum 0.08); dropping the five most influential rows made last-date MAE worse (3.27 to 3.37), so we kept them. Errors track survey noise: 84% of baseline errors are within two SEs of the target, and absolute error correlates with target SE ($r = 0.44$). In the 50–99 band, the three methods in Figure 2 miss by 4.2–5.3 pp on average.

One forward check. The baseline was fixed before release 202620 was published (October 8). On those 71 forecasts its MAE is 3.09, against 3.34 for the six-release mean and 3.90 for carry-forward. One date is not enough to confirm the ranking.

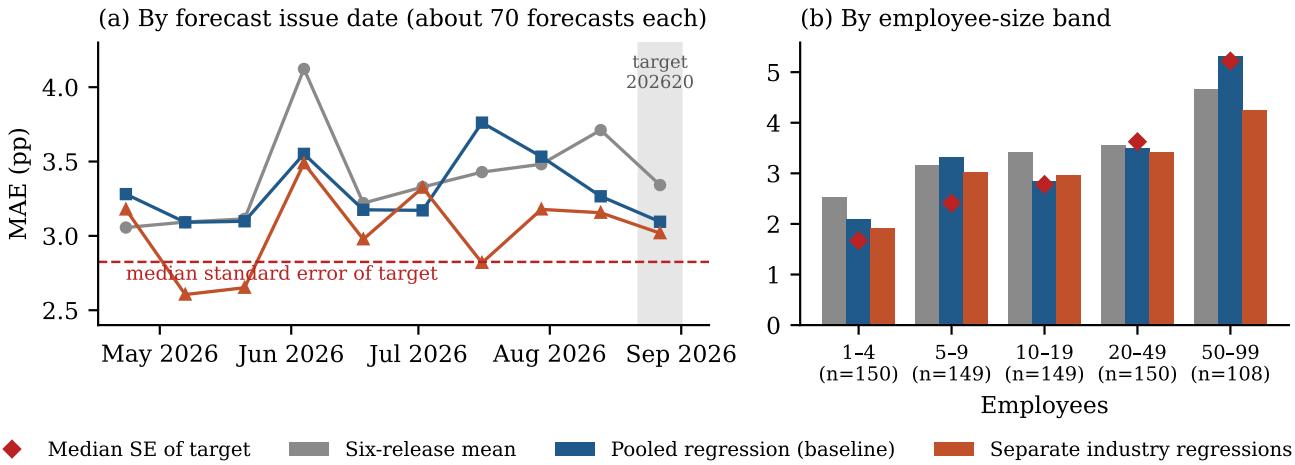


Figure 2. Forecast MAE (a) at each issue date and (b) by size band. Red: median SE of the published target, shown for context. Shaded: forecasts of release 202620, issued August 27.

5. Preliminary answer, limitations, next steps

Answer. Partly. The pooled model's six-week-ahead MAE is 3.30 pp, compared with a median target SE of 2.8 pp. It is not reliably better than a six-release average. Separate industry models look better, but that result is retrospective. Industry differences can inform exploratory discussion; we have not established that forecasts are accurate enough to prioritize outreach, especially for the 50–99 band.

Limitations. (1) Short history: 10 forecast dates with shared shocks, so intervals are wide and fragile. (2) Model selection after outcomes were known. (3) Survey noise, suppression, and screening: results cover 71 well-measured groups, not all small businesses or nonemployers. (4) One downloaded vintage; we cannot prove earlier values were unrevised at each forecast date. (5) Association only: industry differences and adoption trends are not causal.

AI releases and benchmarks (exploratory). Using Epoch AI's benchmark-scored catalog (221 models released January 2024–October 2026), a nonstationary Poisson regression with a time trend and weekday effect was selected by held-out log-likelihood. It expects 12.5 catalogued releases from October 31 to December 11, 2026 (90% simulated range 6–20). Releases arrive in batches more often than the model allows (47 days with two or more releases vs. 26 expected), so that range is too narrow. Adding past release counts, benchmark scores, or expected releases to the adoption forecasts did not reduce error on the same 706 forecasts: +0.13 pp [−0.02, +0.36] for ridge and +0.004 pp [−0.02, +0.03] for boosting, each against a matched control. Benchmark scores were calibrated after the fact, so this test is retrospective. We find no evidence that release timing improves six-week adoption forecasts, and none of this is causal.

Most important for Milestone 3. Freeze the industry model now and score it on releases published after this report, before looking at those outcomes.

6. Team contributions and reflection

Contributions. All five members contributed equally to the project. Ishaan reviewed the proposal; Timothy prepared and submitted the earlier milestone; Chau coordinated meetings and relayed TA feedback. Chris built the industry and nonlinear forecasting extension and the Poisson release model; Rahul audited the workbook, ran early release and benchmark checks, and assembled this report.

What we learned. Aggregate accuracy hid failures: the baseline wins overall but loses in two size bands. Survey uncertainty matters when interpreting errors. Time-ordered evaluation on outcomes not used to choose a model is needed to assess prospective performance.

What failed or changed. Our Milestone 1 hypothesis, that regression beats the six-release mean at most dates and in every size band, was not supported for the pooled baseline. Expected use and business-performance predictors did not help. A regression with every predictor overfit (MAE 6.01). Nonlinear terms gave no gain.

Proposed next-stage task	Owner
Freeze industry model; prospective test on new releases	Chris
Release, benchmark and Poisson work; confirm scope	Rahul
Define the outreach decision and the cost of a forecast miss	Chau
Calibrate forecast intervals by size band	Ishaan and Timothy

AI use. AI tools, including Claude and Codex, helped write and check analysis code, verify results, and organize and edit the report. The team is responsible for every claim.

Sources. U.S. Census Bureau, Business Trends and Outlook Survey, Sector by Employment Size Class workbook and methodology, census.gov/hfp/btos. Epoch AI, Epoch Capabilities Index scores and benchmarks, retrieved October 30, 2026, epoch.ai/eci (CC BY 4.0). Reproducible code accompanies this report.